

Möbius orthogonality and adaptive encoding on a certified Hénon basic set:

exact Parry covariance and a finite-capacity firewall

Bin Wang

August 2026

Abstract

On the certified local survivor of the area-preserving Hénon map $H_6(x, y) = (1 - 6x^2 - y, x)$, we isolate a rigorous typical-exceptional Möbius dichotomy. The survivor is conjugate to a primitive four-state subshift whose scalar sign code forbids two plus signs at distance two. Every fixed periodic point is Möbius-orthogonal for every continuous observable, and Parry-almost every point is simultaneously orthogonal for all continuous observables. In contrast, an explicitly Möbius-adapted point, chosen after the full positive-time arithmetic sequence is read, has raw sign correlation $4/\pi^2$. For the centered variance-one sign observable we prove zero covariance at odd lags, covariance $(-\varphi^{-2})^k$ at lag $2k$, the exact finite-prefix variance formula, and the unconditional bound $V_N \leq \sqrt{5}N$. The density limit $V_N/N \rightarrow 6/\pi^2$ remains conditional on ordinary Cesàro two-point Chowla. The finite-horizon adaptive capacity is exactly two path maximum-weight-independent-set problems, computable in $O(N)$, with $4/\pi^2 \leq \liminf K_N/N \leq \limsup K_N/N \leq 6/\pi^2$. A frozen $N = 2^{20}$ ordering audit returns $p = 0.4111$, a scoped negative finite result. Adaptive encoding is not spontaneous arithmetic coupling, an operator trace, or a zeta bridge. Gates A–E remain false/open.

1 Frozen object and claim boundary

The RH-1–RH-361 foundation is preserved by the four-volume synthesis [8]. RH-364 froze a certified local hyperbolic survivor for the Hénon map and audited a separate engineered prime lift [10]; RH-365 proved a return-bouquet height and radius theorem [9]. Neither result supplies the Möbius statements below, which enter as an independent source-backed edge [11]. The physical route coordinate remains

$$\text{actual_same_clock_unnormalized_head_transport_open.} \quad (1)$$

Consider

$$H_6(x, y) = (1 - 6x^2 - y, x). \quad (2)$$

The frozen geometric input is a locally maximal mixing basic set Λ_* conjugate to the subshift Σ_A with state order $(--, -, +, ++)$ and adjacency matrix

$$A = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix}. \quad (3)$$

This is an inherited theorem, not a conclusion from finite orbit tables [12]. It concerns the local survivor Λ_* , not the full nonwandering set of H_6 .

Writing a state as $w_n = (\varepsilon_n, \varepsilon_{n-1})$, admissibility is equivalent to

$$\varepsilon_n \in \{-1, +1\}, \quad \text{there is no } n \text{ with } \varepsilon_n = \varepsilon_{n+2} = +1. \quad (4)$$

The continuous clipping observable

$$s(x, y) = \begin{cases} -1, & x \leq -1/3, \\ 3x, & |x| < 1/3, \\ +1, & x \geq 1/3 \end{cases} \quad (5)$$

equals the sign code on Λ_* . Indeed the certified survivor lies strictly inside $|x| > 1/3$ on its two branches.

For $z \in \Lambda_*$ and $f \in C(\Lambda_*)$ define

$$\mathcal{C}_N(z, f) = \frac{1}{N} \sum_{n=1}^N \mu(n) f(H_6^n z). \quad (6)$$

This is a time correlation along one orbit. It is not a sum over an orbit catalogue and is not an operator trace.

2 Fixed periodic cancellation and an adaptive exceptional point

Theorem 2.1 (Every fixed periodic orbit is orthogonal). *For every periodic $z \in \Lambda_*$ and every $f \in C(\Lambda_*)$,*

$$\mathcal{C}_N(z, f) \longrightarrow 0. \quad (7)$$

The point, and hence its period, is fixed before $N \rightarrow \infty$.

Proof. If the period is p , then $b_n = f(H_6^n z)$ is a bounded p -periodic sequence. Its finite Fourier expansion is

$$b_n = \sum_{j=0}^{p-1} \widehat{b}_j e^{2\pi i j n / p}.$$

Davenport's uniform estimate gives, for each fixed $A_0 > 0$,

$$\sup_{\theta \in \mathbb{R}} \left| \sum_{n \leq N} \mu(n) e^{2\pi i n \theta} \right| \ll_{A_0} N (\log N)^{-A_0}$$

[1, 3]. Applying it at the finitely many frequencies j/p and dividing by N proves the claim. $\square$

Positive entropy nevertheless gives enough symbolic freedom to encode an arithmetic sequence. Define a bi-infinite sign word by $\varepsilon_n = -1$ for $n \leq 0$, and for $n \geq 1$ set

$$\varepsilon_n = +1 \iff \mu(n) = 1 \text{ and } n \bmod 4 \in \{1, 2\}. \quad (8)$$

Adding two maps the selected residue classes 1, 2 to the unselected classes 3, 0. Thus (4) holds, and the conjugacy supplies a unique point $z_\mu \in \Lambda_*$ with this itinerary.

Theorem 2.2 (Exact adaptive correlation). *For the point selected by (8),*

$$\lim_{N \rightarrow \infty} \mathcal{C}_N(z_\mu, s) = \frac{4}{\pi^2}. \quad (9)$$

Proof. Let $M(N) = \sum_{n \leq N} \mu(n)$. From the definition,

$$\sum_{n \leq N} \mu(n) \varepsilon_n = -M(N) + 2 \# \{n \leq N : \mu(n) = 1, n \bmod 4 \in \{1, 2\}\}. \quad (10)$$

For $r = 1, 2$ one has

$$\frac{1}{N} \sum_{\substack{n \leq N \\ n \equiv r \pmod{4}}} \mu(n) \rightarrow 0, \quad \frac{1}{N} \sum_{\substack{n \leq N \\ n \equiv r \pmod{4}}} \mu(n)^2 \rightarrow \frac{2}{\pi^2}. \quad (11)$$

For $r = 1$, these are the fixed-progression prime number theorem and the squarefree density. For $r = 2$, write $n = 2m$ with m odd. Then $\mu(2m) = -\mu(m)$, while odd squarefree integers have density $4/\pi^2$; counting $m \leq N/2$ again gives $2/\pi^2$ after normalization by N [5]. Since

$$\mathbf{1}_{\{\mu=1\}} = \frac{\mu^2 + \mu}{2},$$

each selected residue class contributes density $1/\pi^2$. Finally $M(N) = o(N)$. Divide (10) by N . $\square$

Remark 2.3 (The essential firewall). The complete positive-time Möbius sequence is read before z_μ is selected. Theorem 2.2 proves offline encoding capacity. It does not prove a nonadaptive prediction rule, a dynamically preferred initial condition, or spontaneous arithmetic coupling. Moreover Λ_* has positive entropy, so this is not a counterexample to Sarnak's zero-entropy conjecture [6]. Positive-entropy systems can display either correlation or uniform decorrelation, and entropy alone is not a switch [4, 2].

3 Parry covariance, variance, and typical orthogonality

Let ν be the Parry measure on Σ_A , transported to Λ_* . Put $\varphi = (1 + \sqrt{5})/2$. In the state order of (3),

$$\pi = \left(\frac{3 + \sqrt{5}}{10}, \frac{1}{5}, \frac{1}{5}, \frac{3 - \sqrt{5}}{10} \right) \quad (12)$$

and

$$P = \begin{pmatrix} \varphi^{-1} & 0 & \varphi^{-2} & 0 \\ 1 & 0 & 0 & 0 \\ 0 & \varphi^{-1} & 0 & \varphi^{-2} \\ 0 & 1 & 0 & 0 \end{pmatrix}. \quad (13)$$

The raw sign has mean $-1/\sqrt{5}$. Hence

$$F(w) = \frac{\sqrt{5}\varepsilon(w) + 1}{2} \quad (14)$$

has mean zero and variance one.

Proposition 3.1 (Exact covariance). *For every $k \geq 0$,*

$$\text{Cov}_\nu(F, F \circ \sigma^{2k+1}) = 0, \quad (15)$$

$$\text{Cov}_\nu(F, F \circ \sigma^{2k}) = (-\varphi^{-2})^k. \quad (16)$$

Proof. As a state vector,

$$F = (-\varphi^{-1}, -\varphi^{-1}, \varphi, \varphi)^\top.$$

Direct multiplication gives

$$P^2 F = -\varphi^{-2} F, \quad \pi \text{diag}(F) P F = 0. \quad (17)$$

Since $\text{Cov}_\nu(F, F \circ \sigma^\ell) = \pi \text{diag}(F) P^\ell F$, the two identities follow. The executable artifact verifies this algebra exactly in $\mathbb{Q}(\sqrt{5})$; no floating eigendecomposition is used. $\square$

For the fixed arithmetic weights define

$$S_N(\omega) = \sum_{n=1}^N \mu(n) F(\sigma^n \omega), \quad V_N = \text{Var}_\nu S_N. \quad (18)$$

Corollary 3.2 (Exact finite-prefix variance). *For every $N \geq 1$,*

$$V_N = \sum_{n \leq N} \mu(n)^2 + 2 \sum_{k \geq 1} (-\varphi^{-2})^k \sum_{n \leq N-2k} \mu(n) \mu(n+2k), \quad (19)$$

where terms with $2k \geq N$ vanish. Unconditionally,

$$0 \leq V_N \leq \sqrt{5} N. \quad (20)$$

If ordinary Cesàro two-point Chowla holds for every fixed even shift, then

$$\frac{V_N}{N} \longrightarrow \frac{6}{\pi^2}. \quad (21)$$

Proof. Expand (18), group pairs by their lag, and insert Proposition 3.1. This gives (19). Taking absolute values in the off-diagonal terms yields

$$V_N \leq N \left(1 + 2 \sum_{k \geq 1} \varphi^{-2k} \right) = N(1 + 2\varphi^{-1}) = \sqrt{5} N.$$

Variance is nonnegative by definition. Under ordinary Chowla, for each fixed k the inner correlation in (19) is $o(N)$. After division by N it is bounded by one, while the coefficients are summable. Dominated convergence therefore passes through the k -sum, and $N^{-1} \sum_{n \leq N} \mu(n)^2 \rightarrow 6/\pi^2$ proves (21). The logarithmically averaged theorem does not imply these ordinary Cesàro hypotheses [7]. $\square$

Theorem 3.3 (Parry-almost-sure simultaneous orthogonality). *For ν -almost every $z \in \Lambda_*$, simultaneously for every $f \in C(\Lambda_*)$,*

$$\mathcal{C}_N(z, f) \longrightarrow 0. \quad (22)$$

Proof. Work first on Σ_A . Let g be a centered locally constant function. Exponential mixing of the finite-state Parry chain gives constants $C_g < \infty$ and $0 < \rho_g < 1$ with

$$|\text{Cov}_\nu(g, g \circ \sigma^h)| \leq C_g \rho_g^h.$$

Because $|\mu(n)| \leq 1$, the weighted partial sums $T_N = \sum_{n \leq N} \mu(n) g \circ \sigma^n$ satisfy $\text{Var}_\nu T_N = O_g(N)$. Chebyshev's inequality at $N = k^2$ gives, for every $\delta > 0$,

$$\sum_{k \geq 1} \nu\{|T_{k^2}| > \delta k^2\} < \infty.$$

Borel–Cantelli yields $T_{k^2}/k^2 \rightarrow 0$ almost surely. Between k^2 and $(k+1)^2$ there are at most $2k+1$ additional bounded summands, so convergence holds for every N .

Choose a countable uniformly dense family of rational-valued locally constant functions. Intersect their full-measure convergence sets. For a noncentered function the constant component contributes a multiple of $M(N)/N$, which vanishes by the prime number theorem. Uniform approximation, together with $N^{-1} \sum_{n \leq N} |\mu(n)| \leq 1$, extends the conclusion on this one full-measure set to every continuous function. Transport through the conjugacy. $\square$

The quantifier is important: one full-measure set works for all continuous observables, but the conclusion is not uniform over all $z \in \Lambda_*$ because Theorem 2.2 supplies an explicit exception.

4 Finite-horizon adaptive capacity

Definition 4.1. For $N \geq 1$, define the open raw capacity

$$K_N = \max_{\substack{\varepsilon_1, \dots, \varepsilon_N \in \{-1, +1\} \\ \varepsilon_n = \varepsilon_{n+2} = +1 \text{ never}}} \left| \sum_{n=1}^N \mu(n) \varepsilon_n \right|. \quad (23)$$

The optimizing word may depend on N and on the complete Möbius prefix.

Theorem 4.2 (Linear-time exact optimization and asymptotic bracket). *The value K_N and a witnessing word can be computed exactly in $O(N)$ time. Moreover*

$$\frac{4}{\pi^2} \leq \liminf_{N \rightarrow \infty} \frac{K_N}{N} \leq \limsup_{N \rightarrow \infty} \frac{K_N}{N} \leq \frac{6}{\pi^2}. \quad (24)$$

Proof. Start from the all-negative word. If I is the set of positions changed to plus, then

$$\sum_{n \leq N} \mu(n) \varepsilon_n = -M(N) + 2 \sum_{n \in I} \mu(n). \quad (25)$$

The constraint says exactly that I is an independent set in the disjoint union of the odd-index path and the even-index path. For path weights a_j , the maximum-weight-independent-set recurrence is

$$D_j = \max\{D_{j-1}, D_{j-2} + a_j\}, \quad D_0 = 0, \quad D_1 = \max\{0, a_1\}. \quad (26)$$

Run it on both paths with weights $\mu(n)$ to maximize (25), and with weights $-\mu(n)$ to minimize it. Traceback recovers witnesses. The total work is linear.

Every prefix of the word in (8) is feasible, so Theorem 2.2 gives the lower bound in (24). For every sign word,

$$\left| \sum_{n \leq N} \mu(n) \varepsilon_n \right| \leq \sum_{n \leq N} \mu(n)^2.$$

The squarefree density gives the upper bound. $\square$

No existence theorem for $\lim K_N/N$ follows. The optimizer can change with N , so a nonvanishing envelope also does not contradict Theorem 2.1, which fixes one periodic point before taking the limit. Filling signs outside $[1, N]$ with -1 realizes each open optimizer by an N -dependent point of Λ_* .

5 Frozen finite audit and numerical boundary

The source experiment fixed $N_{\max} = 2^{20}$, 1024 stationary Parry paths, and 1023 conditional arithmetic surrogates before reading the production results [11]. The surrogates preserve the squarefree zero support and the numbers of positive and negative signs within frozen growing blocks; every surrogate undergoes the same adaptive optimization. The rank statistic is

$$p_{\text{cap}} = \frac{1 + \#\{b : K_{N_{\max}, b} \geq K_{N_{\max}, \text{real}}\}}{1024}. \quad (27)$$

The rank test is a scoped negative result: the natural ordering is not exceptional under this frozen finite null. It is not a probability model for Möbius itself, and it does not prove any asymptotic ordering law.

The variance distinction is equally strict. Formula (19) is exact. The upstream production routine, however, evaluates it in floating arithmetic and stops after the geometric coefficient falls

Table 1: Locked endpoint diagnostics. These are finite reproduction facts, not asymptotic theorems.

Quantity at $N = 2^{20}$	Value
Exceptional raw integer score	425025
Exceptional raw correlation	0.405335426
Open maximum / minimum	515983 / - 516163
Open absolute capacity	0.492251396
Conditional-null exceedances	420 of 1023
Rank p -value	421/1024 = 0.4111328125
Stored floating variance density	0.608082565

below 10^{-15} . The first omitted geometric tail has normalized absolute bound below 1.3×10^{-15} before floating roundoff, but the stored decimal is still a tolerance-controlled diagnostic, not an exact algebraic evaluation. The RH-366 artifact instead checks small finite prefixes exactly in $\mathbb{Q}(\sqrt{5})$.

The executable package also performs the following independent checks:

1. all frozen source and Paper-3 dependency hashes;
2. exhaustive equivalence of the sign rule and graph cycles through length eight;
3. the exceptional integer identity on finite prefixes;
4. exact $\mathbb{Q}(\sqrt{5})$ covariance and variance rows;
5. dynamic programming against brute force through length twelve;
6. the four-volume replay, strict JSON parsing, fixed archive membership, and mutation rejection.

Finite checks reproduce identities and provenance. They are not evidence for the prime number theorem, ordinary Chowla, or a capacity limit.

6 Route verdict and Gate ledger

Table 2: The strongest warranted route interpretation.

Route	Verdict
Route A	GO . Fixed periodic and Parry-typical orthogonality, the exact adaptive correlation, covariance and variance theorems, the linear-time capacity theorem, and the frozen scoped negative form a standalone theorem package.
Route B	STOP_SCOPED . The first fatal mismatch is data type: z_μ is chosen after reading μ . Its scalar time correlation is not a canonical determinant, operator trace, or signed von-Mangoldt prime-power ledger.

The following boundaries are mandatory.

1. Λ_* is a certified local basic set, not the full Hénon dynamics.
2. The exceptional point is an offline encoding, not an intrinsic or independently selected coupling.
3. The Parry theorem is almost sure and simultaneous in f , not uniform over all points or measures.

4. The variance-density limit is conditional on ordinary fixed-shift Cesàro Chowla.
5. The finite capacity plateau is not a proved limit.
6. Möbius-weighted orbit averages are not von-Mangoldt traces and do not define an Euler product or an L-function family.

Consequently the Gate ledger remains

Gate	Status	First missing object
A	false/open	canonical intrinsic dynamical spectral determinant
B	false/open	time-oriented scattering or unitary completion
C	false/open	self-adjoint generator and intrinsic $T \log T$ law
D	false/open	signed von-Mangoldt prime-power operator trace
E	false/open	completed-zeta divisor equality

No Hilbert–Pólya operator, Riemann-zero spectral identification, completed-zeta divisor equality, or proof of the Riemann Hypothesis follows.

7 Conclusion and next exact questions

One explicit positive-entropy Hénon subsystem supports both rigorous Möbius cancellation and rigorous arithmetic tracking. The distinction is selection: fixed periodic and Parry-typical points cancel, while a point chosen after reading the arithmetic sequence tracks it with correlation $4/\pi^2$. Exact covariance and linear-time capacity quantify the mechanism without promoting it to a spectral or zeta construction.

The shortest new theorem problems are correspondingly narrow: determine whether K_N/N has a limit; classify nonadaptive invariant measures with quantitative Möbius cancellation; and generalize the graph-capacity theorem to other mixing subshifts. Independent breadth-first routes, including the cyclic-Ulam structural audit and the RH-365 primitive-divisor/radius problems, remain separate. Repackaging adaptive encoding as intrinsic coupling would not be a continuation.

References

- [1] Harold Davenport. On some infinite series involving arithmetical functions (II). *The Quarterly Journal of Mathematics*, os-8(1):313–320, 1937.
- [2] Tomasz Downarowicz and Jacek Serafin. Almost full entropy subshifts uncorrelated to the Möbius function. *International Mathematics Research Notices*, 2019(11):3459–3472, 2019.
- [3] Henryk Iwaniec and Emmanuel Kowalski. *Analytic Number Theory*, volume 53 of *American Mathematical Society Colloquium Publications*. American Mathematical Society, Providence, RI, 2004.
- [4] D. Karagulyan. On Möbius orthogonality for subshifts of finite type with positive topological entropy. *Studia Mathematica*, 237(3):277–282, 2017.
- [5] Leon Mirsky. Arithmetical pattern problems relating to divisibility by r th powers. *Proceedings of the London Mathematical Society*, s2-50(1):497–508, 1948.
- [6] Peter Sarnak. Three lectures on the Möbius function, randomness and dynamics. Institute for Advanced Study lecture notes, 2010.

- [7] Terence Tao. The logarithmically averaged Chowla and Elliott conjectures for two-point correlations. *Forum of Mathematics, Pi*, 4:e8, 2016.
- [8] Bin Wang. Conditional prime dynamics: A corpus frontier synthesis. Four-volume repository synthesis of RH-1–RH-361, 2026.
- [9] Bin Wang. Prime-return bouquet height and analytic radius: Midpoint compression and prime-order anchors. RH-365 repository paper, 2026.
- [10] Bin Wang. Weighted Hénon prime lifts: Analytic domains, fredholm regions, and a cubic trace obstruction. RH-364 repository paper, 2026.
- [11] Liang Wang. Hénon Möbius correlations: Periodic and typical orthogonality versus adaptive encoding. Repository preprint, commit 34490443f50cfe9af9ff93888e51e7e7e534a5a7, 2026.
- [12] Liang Wang. Weighted cycles and finite-resolution operators on an explicit hyperbolic Hénon survivor. Repository preprint, commit ff44f961261349848c9f65ede6a031b7e155aca9, 2026.